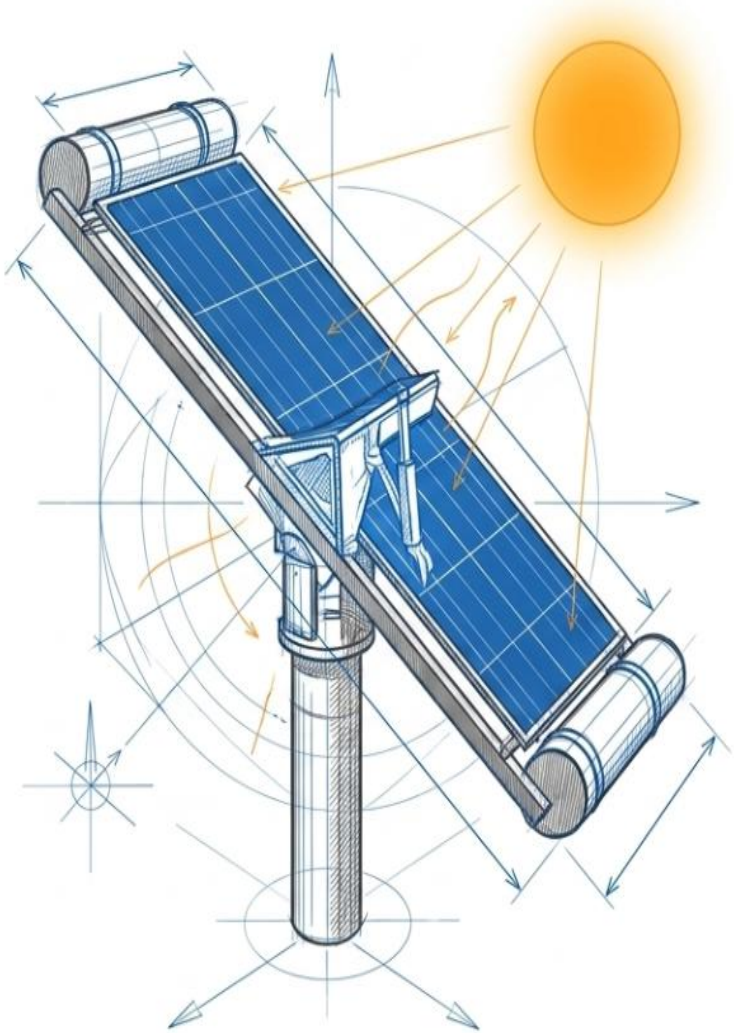


Design and Development of a Passive Solar Tracking System

ME325 — Third Year Engineering Project Proposal | Academic Year 2026

Team Name – SolarTrack

Group Members: E/21/114 · E/21/387 · E/21/446

Supervisor: Ms. L.U. Bakmeedeniya

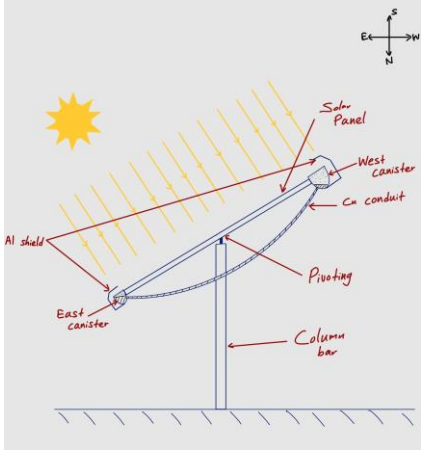
The Need & Aim

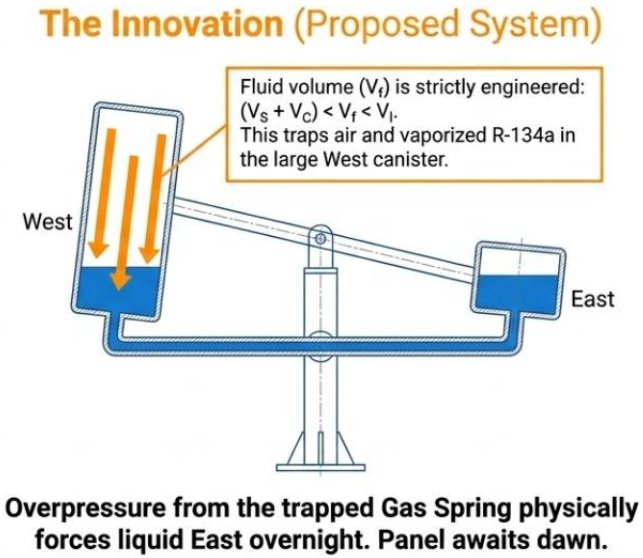
- Fixed solar panels lose up to 35% of daily energy potential & active trackers consume 2–5% of output.

Feature	Active Tracker	Zomeworks Passive	Our System
Tracks the Sun	✓	✓	✓
Power Consumption	2–5% of output	None	None
Morning Reset	Reliable	Unreliable	Reliable
Maintenance	High	Minimal	Minimal
Suitable for Rural	✗	✓	✓

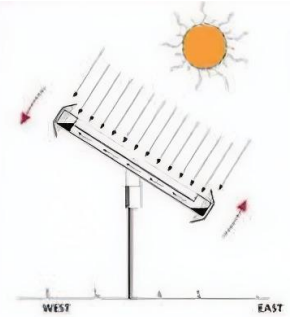
- **The Aim :** Design passive solar tracker using differential vapor pressure and gas spring mechanism — zero electrical power, reliable daily reset under tropical conditions.

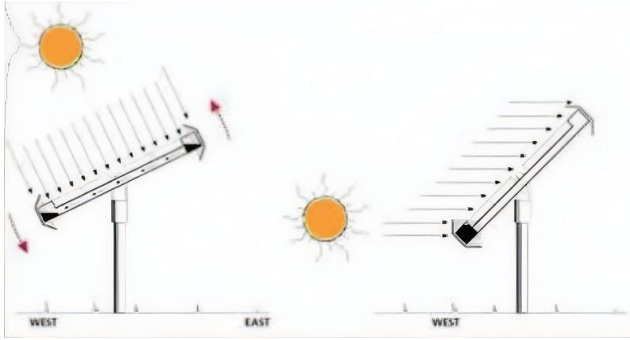
Engineering Design & Methodology





Overpressure from the trapped Gas Spring physically forces liquid East overnight. Panel awaits dawn.





Software

Modeling

Thermodynamic Simulation

Engineering Drawings

Structural & Thermal FEA

Project Timeline

Task \ Week	1	2	3	4	5	6	7	8	9	10	11	12
Research and Conceptualization	■	■	■									
Detailed Design & Analysis			■	■	■	■						
Simulation/ Prototyping/ Verification					■	■	■	■	■	■		
Final Evaluation & Documentation											■	■

The Big Three Impact

💰 Economic Impact

- 40–60% lower cost than active trackers
- Near-zero operational cost
- 14.5% energy improvement (704 vs 614.8 W/m²)
- Locally manufacturable

👥 Social Impact

- Deployable in rural & off-grid communities
- Zero maintenance electronics
- Improves energy access
- Educational demonstration

🌿 Environmental Impact

- Zero operational emissions
- Eliminates e-waste
- R-134a working fluid — ozone-safe
- Supports 70% renewable target by 2030